Updating a Scoping Review of Research Data Sharing with Machine Learning Methods

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Aim

We are updating the 2019 review of research data sharing (Chawinga and Zinn 2019) with scoping review methods from Cavallaro et al. (2026). The 2019 review matches our focus on research data sharing, which makes it a strong candidate for an update. Where it looks at differences across countries, we look at differences across research domains, with special attention on healthcare. Our method draws on the 2026 protocol, which pairs machine learning screening with automated deduplication, and improves on it in two places: deduplication and seed generation.

Frameworks and Tools

We follow the PRISMA extension for scoping reviews (PRISMA-ScR) and the Joanna Briggs Institute scoping review (JBI) methodology. Both are standard for scoping reviews, make the work reproducible, and are supported by a broad ecosystem of tools and resources.

For deduplication we use ASySD (also used by Perrier et al. (2017)), that, in contrast with Zotero workflow used by Cavallaro et al. (2026), is scriptable for reproducibility purposes. ASReview uses machine learning to rank records by predicted relevance, leveraging Naive Bayes text classification. It needs a seed sample to initialize the ranking. Cavallaro et al. (2026) select their seed sample ad-hoc from the first 5% of deduplicated records.

Search and Seed

All four queries build on Chawinga and Zinn (2019). The base query takes their core term, "research data sharing" (RDS). The three concept queries extend that term with the vocabulary the field uses around it. Each pairs one key term (open data, data governance or open science) with "research data" or "data sharing" using AND, so a record must name both the concept and the data practice in its title or abstract. All four queries cover 2015 to 2025, which brackets the 2019 review and picks up where it left off. We ran them in PubMed, Scopus and Web of Science, translating field tags, date limits and document types for each database, and the appendix lists the full strings.

The base RDS query alone serves as the seed. It returned 308 records against 4,192 for the four queries combined (7.3%), modestly above the conventional 5%. A seed derived from a fixed keyword is reproducible, and its size stays near where a seed sample should land.

Table 1: Records retrieved per query and database, 2015 to 2025, before deduplication. Percentages give each query's share of the database's summed hits.

Query	PubMed N (% of col.)	Scopus N (% of col.)	WoS N (% of col.)
Q1 research data sharing	64 (6.1%)	135 (7.8%)	109 (7.7%)
Q2 open data	352 (33.6%)	644 (37.4%)	564 (39.7%)
Q3 data governance	219 (20.9%)	348 (20.2%)	255 (18.0%)
Q4 open science	414 (39.5%)	596 (34.6%)	492 (34.6%)
Column total	**1,049**	**1,723**	**1,420**

Note: All terms searched in titles and abstracts. Database-specific strings appear in the appendix. Results presented above have not yet been deduplicated.

Q1: `"research data sharing" OR "research data-sharing"`
Q2: `"open data" AND ("research data" OR "data sharing")`
Q3: `"data governance" AND ("research data" OR "data sharing")`
Q4: `"open science" AND ("research data" OR "data sharing")`

We also checked candidate terms from earlier drafts of the query against the seed. Table 3 shows how often each term appears in the records the base RDS query returned, which guided the four queries above.

Rank	Term	Total	PubMed	WoS	Scopus

Table 3: Candidate terms ranked by records matched across the three databases. Ties share a rank (=).

Rank	Term	Total	PubMed	WoS	Scopus
1	policies	88	16	31	41
2	challenge*	73	21	23	29
3	policy	69	14	22	33
4	repositor*	68	15	20	33
5	ethic*	62	19	20	23
6	privacy	52	15	15	22
7	infrastructure	46	9	14	23
8	barrier*	45	10	16	19
9	perception*	42	9	15	18
10	open science	41	5	15	21
11	motivat*	37	5	15	17
12	interoperab*	36	10	11	15
13	“open data”	34	7	12	15
14	open access	32	3	9	20
15	consent	31	9	10	12
16	open research	27	6	9	12
17	licens*	25	3	5	17
18	attitude*	23	3	9	11
19	incentive*	21	3	6	12
20	mandate*	17	4	6	7
21	willingness	15	3	5	7
22	obstacle*	12	4	3	5
23=	“data reuse”	10	0	4	6
23=	stewardship	10	3	3	4
25	“data management plan*”	9	2	3	4
26=	“data curation”	8	2	3	3
26=	facilitator*	8	2	3	3
28	“data re-use”	5	1	1	3
29	enabler*	4	1	1	2
30	reluctan*	3	1	1	1
31	“digital curation”	0	0	0	0

Manual annotation of seed sample

Terms from the second block of the prior draft query were implicitly grouped around five themes:

- Factors that obstruct
- Factors that facilitate
- Stakeholder perspectives

- Policies and what drive them
- Mechanisms for implementation

We repurposed these themes so that they may inform the screening strategy for annotating the seed.

Tools and Frameworks

- PRISMA-ScR: https://www.prisma-statement.org/scoping
- JBI scoping review methodology: https://doi.org/10.46658/JBIMES-24-09
- ASySD: https://github.com/camaradesuk/ASySD
- ASReview: https://asreview.nl
- Zotero: https://www.zotero.org

References

Cavallaro, Ludovico, Philipp Graf, Josephine Verona Konrad, et al. 2026. "Conceptualization, Operationalization, and Measurement of Adherence to Patient-Facing Digital Health Applications: Protocol for a Scoping Review." *JMIR Research Protocols* 15: e95287. https://doi.org/10.2196/95287.

Chawinga, Winner Dominic, and Sandy Zinn. 2019. "Global Perspectives of Research Data Sharing: A Systematic Literature Review." *Library & Information Science Research* 41 (2): 109–22. https://doi.org/10.1016/j.lisr.2019.04.004.

Perrier, Laure, Erik Blondal, A. Patricia Ayala, et al. 2017. "Research Data Management in Academic Institutions: A Scoping Review." *PLOS ONE* 12 (5): e0178261. https://doi.org/10.1371/journal.pone.0178261.

Appendix: Full Search Queries

Generic Queries

Database-neutral core of the four queries. Each database adds its own field tags, date limits and document type filters. All terms are searched in titles and abstracts.

```
-- Q1: research data sharing (base RDS query)
"research data sharing" OR "research data-sharing"

-- Q2: open data
"open data"
AND ("research data" OR "data sharing")

-- Q3: data governance
"data governance"
AND ("research data" OR "data sharing")

-- Q4: open science
"open science"
AND ("research data" OR "data sharing")
```

Scopus (1,723 records)

```
-- Q1 (135 records)
TITLE-ABS ( "research data sharing" OR "research data-sharing" )
AND PUBYEAR > 2014 AND PUBYEAR < 2026
AND DOCTYPE ( ar OR re )

-- Q2 (644 records)
TITLE-ABS ( "open data" )
AND TITLE-ABS ( "research data" OR "data sharing" )
```

```
AND PUBYEAR > 2014 AND PUBYEAR < 2026
AND DOCTYPE ( ar OR re )

-- Q3 (348 records)
TITLE-ABS ( "data governance" )
AND TITLE-ABS ( "research data" OR "data sharing" )
AND PUBYEAR > 2014 AND PUBYEAR < 2026
AND DOCTYPE ( ar OR re )

-- Q4 (596 records)
TITLE-ABS ( "open science" )
AND TITLE-ABS ( "research data" OR "data sharing" )
AND PUBYEAR > 2014 AND PUBYEAR < 2026
AND DOCTYPE ( ar OR re )
```

Web of Science (1,420 records)

```
-- Q1 (109 records)
(
  TI=( "research data sharing" OR "research data-sharing" )
  OR AB=( "research data sharing" OR "research data-sharing" )
)
AND PY=(2015-2025)
AND DT=(Article OR Review)

-- Q2 (564 records)
(
  TI=( "open data" )
  OR AB=( "open data" )
)
AND
(
  TI=( "research data" OR "data sharing" )
  OR AB=( "research data" OR "data sharing" )
)
AND PY=(2015-2025)
AND DT=(Article OR Review)

-- Q3 (255 records)
(
  TI=( "data governance" )
  OR AB=( "data governance" )
)
AND
(
  TI=( "research data" OR "data sharing" )
  OR AB=( "research data" OR "data sharing" )
)
AND PY=(2015-2025)
AND DT=(Article OR Review)

-- Q4 (492 records)
(
  TI=( "open science" )
  OR AB=( "open science" )
```

```
)
AND
(
  TI=( "research data" OR "data sharing" )
  OR AB=( "research data" OR "data sharing" )
)
AND PY=(2015-2025)
AND DT=(Article OR Review)
```

PubMed (1,049 records)

```
-- Q1 (64 records)
( "research data sharing"[tiab] OR "research data-sharing"[tiab] )
AND 2015:2025[dp]
AND "Journal Article"[pt]

-- Q2 (352 records)
"open data"[tiab]
AND ( "research data"[tiab] OR "data sharing"[tiab] )
AND 2015:2025[dp]
AND "Journal Article"[pt]

-- Q3 (219 records)
"data governance"[tiab]
AND ( "research data"[tiab] OR "data sharing"[tiab] )
AND 2015:2025[dp]
AND "Journal Article"[pt]

-- Q4 (414 records)
"open science"[tiab]
AND ( "research data"[tiab] OR "data sharing"[tiab] )
AND 2015:2025[dp]
AND "Journal Article"[pt]
```